

Job Posting: 178573 - Position: S26 Digital Twin Development Assistant (Back End Infrastructure) 178573

Co-op Work Term Posted: 2026 - Summer
App Deadline 03/02/2026 09:00 AM
Application Method: Through UBC Science Co-op
Posting Goes Live: 02/23/2026 02:32 PM
Job Posting Status: Approved

ORGANIZATION INFORMATION

Organization UBC Mechanical Engineering
Address Line 1 2360 East Mall
City Vancouver
Postal Code / Zip Code V6T 1Z3
Province / State British Columbia
Country Canada

JOB POSTING INFORMATION

Placement Term 2026 - Summer
** Job Title ** S26 Digital Twin Development Assistant (Back End Infrastructure) 178573
Position Type Co-op Position
Job Location Vancouver, BC
Country Canada
Duration 4 months
Salary Currency CAD
Salary 30.0 per hour for 0 Major List
Job Description

Period: May-August 2026 (4 months) with the possibility of extension for an additional term based on project needs and student performance

Reports To: The Digital Twin Development Assistant will report to MéridaLabs Managers and Researchers

Job Overview:

The MéridaLabs research group in the UBC Department of Mechanical Engineering is seeking a co-op student to assist with tasks related to development of the digital twin for the Smart Hydrogen Energy District for the period of May-August 2026, with the possibility of extension for the September-December 2026 term dependent on project needs and student performance.

Specifically, the group is seeking to further develop and refine its Digital Twin model of the Smart Hydrogen Energy District, to provide real-time and historical visualization of energy generation, storage, distribution, and building systems components. The digital twin will eventually be displayed internally and to invite-only external stakeholders on a large display in the MéridaLabs control room.

The desired outcomes of this co-op work are as follows:

- Implementation or refinement of real-time data integration pipelines (e.g., OPC UA, MQTT, API-based ingestion)
- Deployment or configuration of on-prem dashboard systems (e.g., Grafana, InfluxDB, or related tools)
- Documentation of system architecture, deployment steps, and configuration for long-term maintainability
- Support for cloud-based Digital Twin components (e.g., AWS services for data storage, visualization, or analytics)

Requirements and Qualifications. Please be as detailed as possible.

Responsibilities:

- Assist in configuring and deploying on-prem Digital Twin infrastructure
- Support integration of data sources (OPC UA, MQTT, REST APIs)
- Configure and maintain time-series databases (e.g., InfluxDB) and dashboard tools (e.g., Grafana)
- Assist with AWS-based deployment and cloud architecture components
- Support containerization and environment setup (Docker, Linux-based systems)
- Contribute to cybersecurity and access control configuration (e.g., SSO integration)

Job Requirements**Qualifications/Preferred Skills:**

- Linux environment familiarity
- Experience with Docker or containerization
- Basic networking knowledge (ports, firewalls, client/server architecture)
- Familiarity with cloud services (AWS preferred)
- Experience with databases (SQL or time-series)
- Programming experience (C++/ JAVA/ Python)

Citizenship Requirement N/A

APPLICATION INFORMATION

Application Procedure	Through UBC Science Co-op
Cover Letter Required?	Yes
Address Cover Letter to	Hiring Manager